

Proposition Logic

26 June 2026 05:18

Proposition were given statements that results either 'True' or 'False'.
Consider variables A, B, C,... which can have two values either True or False.

A = "Anchit is going to school" $\begin{cases} \rightarrow \text{true} \\ \rightarrow \text{false} \end{cases}$

There are 3 logical operators : $\begin{cases} \neg & \text{not} \\ \wedge & \text{and} \\ \vee & \text{or} \end{cases}$

Negation or complement : $\text{not}(\text{not } A) = A$, $\text{not}(1) = 0$, $\text{not}(0) = 1$
 $\text{not } A = \bar{A}$, $\text{not}(\bar{A}) = A$

Associative : $(A \text{ and } B) \text{ and } C = A \text{ and } (B \text{ and } C)$ $\quad \quad \quad \begin{cases} '\wedge' \leftarrow \text{and} \\ '\vee' \leftarrow \text{or} \end{cases}$
 $(A \wedge B) \wedge C = A \wedge (B \wedge C)$
 $(A \text{ or } B) \text{ or } C = A \text{ or } (B \text{ or } C)$
 $(A \vee B) \vee C = A \vee (B \vee C)$

would you like a coke $\vee$ orange juice = $(A \vee B)$
 $\quad \quad \quad \uparrow$
 $\quad \quad \quad \text{operator}$

I will not either take coke or orange juice = $(A \vee B)'$

OR = ' $\vee$ '
AND = ' $\wedge$ '
NOT = ' $\neg$ ', '-', ' $\neg$ '

$\neg A$

X = sky is clear

$\bar{X}$ = sky is not clear
 $\quad \quad \quad \nwarrow$ operator

Implication : 'X' is true and 'Y' is false, then the result is 'false' or otherwise it is true. It is represented by ' $\Rightarrow$ ' or ' $\rightarrow$ '.
if...then.

	X	Y	$X \Rightarrow Y$
0	0	0	1
1	0	1	1

	X	Y	$X \Rightarrow Y$
0	0	0	1
1	0	1	1
2	1	0	0
3	1	1	1

✓ POS
 $X \Rightarrow Y = \overline{X} + Y$

$X = \text{sky is clear}$
 $Y = \text{It is raining}$
 $X \Rightarrow Y = \text{if sky is not clear then it is raining}$

$Y \Rightarrow X = \text{if It is not raining then sky is clear.}$
 operator
 operands

A = "You practice for exam"

B = "You will succeed"

"We need to write maths logic"

$A \rightarrow B = \text{if not you practice for exam, then you will succeed.}$

if A then B = $A \Rightarrow B = (\overline{A} + B)$ * Algebraic form

if $\frac{5=7}{F}$ then $\frac{15=22}{F} \Rightarrow \text{True}$

if $\frac{x=2}{T}$ then $\frac{2x=4}{T} \Rightarrow \text{True}$

if $\frac{x=2}{\text{True}}$ then $\frac{3x=5}{\text{Not possible}} \Rightarrow \text{False}$

if $\frac{x \neq 2}{\text{true}}$ then $\frac{3x=5}{\downarrow \text{True}} \Rightarrow \text{True}$

if $\frac{5x > 15}{\text{False}}$ then $\frac{5x = 15}{\text{True}} \Rightarrow \text{True} \quad [\because x = 3]$

Equivalence: Both statement or variable have similar values to result into true; it is called if and only if or iff.
 It is represented by $\Leftrightarrow$, or $\Leftrightarrow$

	X	Y	$X \Leftrightarrow Y$
0	0	0	1
1	0	1	0
2	1	0	0
3	1	1	1

$\overline{X} \cdot \overline{Y}$
 $\{ (X + \overline{Y})$
 $\{ (\overline{X} + Y)$
 XY

$$\text{SOP} = X \Leftrightarrow Y = \overline{X}\overline{Y} + XY$$

$$\text{POS} = (X \Leftrightarrow Y) = (X + \overline{Y})(\overline{X} + Y)$$

$X = \text{sky is clear}$
 $Y = \text{It is raining}$

$X \leftrightarrow Y = \text{if sky is clear only if it is not raining}$
 $Y \leftrightarrow X = \text{if it is raining only if sky is not clear.}$

Contrapositive: $A \rightarrow B$ and $(\text{not } B) \rightarrow (\text{not } A) =$

$$\begin{aligned}
 & (\bar{A} + B) \cdot ((\bar{B})' + \bar{A}) \\
 &= (\bar{A} + B) (B + \bar{A}) \\
 &= \underline{\bar{A} + \bar{A}B} + B + \bar{A}B \\
 &= \bar{A} (1 + B) + B (1 + \bar{A}) \\
 &= \bar{A} + B
 \end{aligned}$$

A	B	A'	B'	A → B	B → A
0	0	1	1	1	1
0	1	1	0	1	1
1	0	0	1	0	0
1	1	0	0	1	1

↑ = ↑

This contra-positive.

Tautology: - When the resulting column have nothing but only '1'

	X	Y	logic
0	0	0	1
1	0	1	1
2	1	0	1
3	1	1	1

Contradiction - When the resulting column have nothing but only '0'

	X	Y	logic
0	0	0	0
1	0	1	0
2	1	0	0
3	1	1	0

Inverse :

Converse :

antecedent consequent
 $P \rightarrow Q \equiv \bar{P} \rightarrow \bar{Q}$

$P \rightarrow Q \equiv Q \rightarrow P$

Consistent statement is a logical statement which always generate 'true' or 'false' value as per given logic. For example: $A \wedge B$, $A \vee B$

De-Morgan's law (i) $(P \wedge Q)' = \bar{P} \vee \bar{Q}$

De-Morgan's law (i) $(p \wedge q)' = \bar{p} \vee \bar{q}$
(ii) $(p \vee q)' = \bar{p} \wedge \bar{q}$

H.W Draw Truth Table to prove De'Morgan's law.

'+' $\leftarrow$ conjunction

'.' $\leftarrow$ disjunction.

same $\swarrow$ correct technical word.

In propositional logic, a contingency is a compound proposition that is true in some cases and false in others, and therefore is neither a tautology nor a contradiction.

From <<https://en.flamath.com/logical-contingency>>